

ML Assignment

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Repository: <https://github.com/vicky150612/ML-Assignment-1>

1 Introduction

The objective of this assignment is to build polynomial regression models for the given geothermal energy datasets. Two problems were considered:

- **Phase 1:** Power Plant Steam Turbine Optimization
- **Phase 2:** Subterranean Thermal Reservoir Mapping

For both phases, polynomial models of different degrees were evaluated using five-fold cross-validation. Ordinary Least Squares (OLS), Ridge, and Lasso regression were considered. The main evaluation metrics were Mean Squared Error (MSE) and R^2 .

2 Common Implementation

The training data was loaded using `pandas`, and the feature columns were separated from the target variable. Polynomial features were generated using `PolynomialFeatures` with `include_bias=False`, so the constant term was handled separately.

For every polynomial degree, the resulting features were standardized using `StandardScaler`. During cross-validation, the scaler was fitted only on the training portion of each fold and then applied to the validation portion. This prevents information from the validation set from affecting the transformation.

Five-fold cross-validation was performed with shuffled folds using `random_state=42`. For every fold, the average validation MSE and R^2 were calculated for each candidate model.

Regularisation was used to control the effect of large numbers of polynomial terms and to improve stability at higher degrees. Ridge applies L_2 regularisation, while Lasso applies L_1 regularisation. The regularised Ridge objective can be written as

$$\min_{\theta} \left(\|y - X\theta\|_2^2 + \alpha \|\theta\|_2^2 \right),$$

while Lasso uses an L_1 penalty:

$$\min_{\theta} \left(\|y - X\theta\|_2^2 + \alpha \|\theta\|_1 \right).$$

For OLS, the coefficients were obtained using singular value decomposition rather than directly inverting $X^T X$. Very small singular values were discarded using the threshold

$$s_i > 10^{-10} s_{\max},$$

which helps avoid numerical instability.

For Ridge, the solution was calculated directly using the regularised normal equation. Lasso was evaluated using a Lasso path over a predefined set of α values.

For each polynomial degree, the best α for Ridge and Lasso was selected using the lowest validation MSE. Finally, the overall model was selected using the lowest cross-validation MSE, with R^2 used as a tie-breaker. The selected model was then retrained on the complete training dataset before predicting the test data.

3 Phase 1: Steam Turbine Optimization

3.1 Implementation

Phase 1 contains six input features,

$$x_1, x_2, x_3, x_4, x_5, x_6,$$

and polynomial degrees from 1 to 10 were tested.

OLS, Ridge, and Lasso were compared for every degree. For Ridge, 11 values of α were tested using a logarithmic range from

$$10^{-2} \text{ to } 10^3,$$

and for Lasso the following six values were used:

$$\alpha \in \{0.1, 0.05, 0.02, 0.01, 0.005, 0.002\}.$$

The Lasso path used a maximum of 3000 iterations and a tolerance of 10^{-3} during cross-validation.

3.2 Model Selection

The cross-validation results are shown in Figure 1.

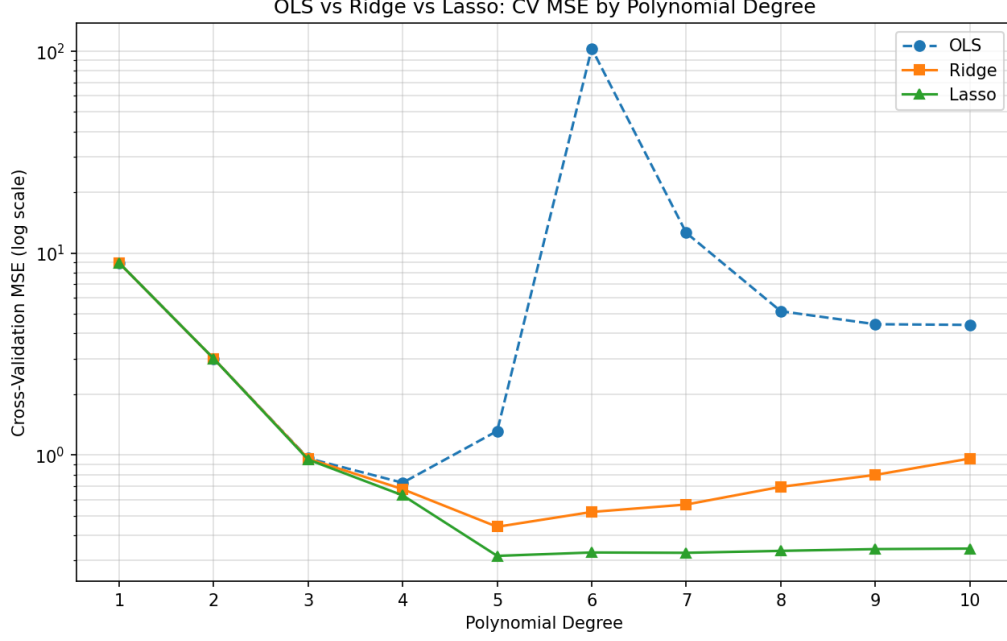


Figure 1: OLS, Ridge and Lasso cross-validation MSE for different polynomial degrees.

At lower degrees, the three methods have similar performance. As the degree increases, OLS becomes increasingly unstable, while regularised methods remain more stable.

The best model was Lasso with degree 5 and

$$\alpha = 0.01$$

with

$$\text{CV MSE} = 0.316115, \quad R^2 = 0.966992.$$

For the final training step, the selected Lasso model was fitted using the complete training dataset. The Lasso path was run with a maximum of 20000 iterations and a tolerance of 10^{-5} .

The final training performance was

$$\text{MSE} = 0.236171, \quad R^2 = 0.976029.$$

The trained model was then used to generate predictions for the test dataset.

4 Phase 2: Subterranean Thermal Reservoir Mapping

4.1 Implementation

Phase 2 contains three input features,

$$x_1, \quad x_2, \quad x_3,$$

and polynomial degrees from 1 to 20 were tested.

OLS, Ridge, and Lasso were again evaluated using the same five-fold cross-validation procedure. For Ridge, 12 values of α were tested using a logarithmic range from

$$10^{-3} \text{ to } 10^{2.5},$$

while Lasso used

$$\alpha \in \{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}\}.$$

During cross-validation, the Lasso path used a maximum of 3000 iterations and a tolerance of 10^{-3} .

4.2 Model Selection

The cross-validation results are shown in Figure 2.

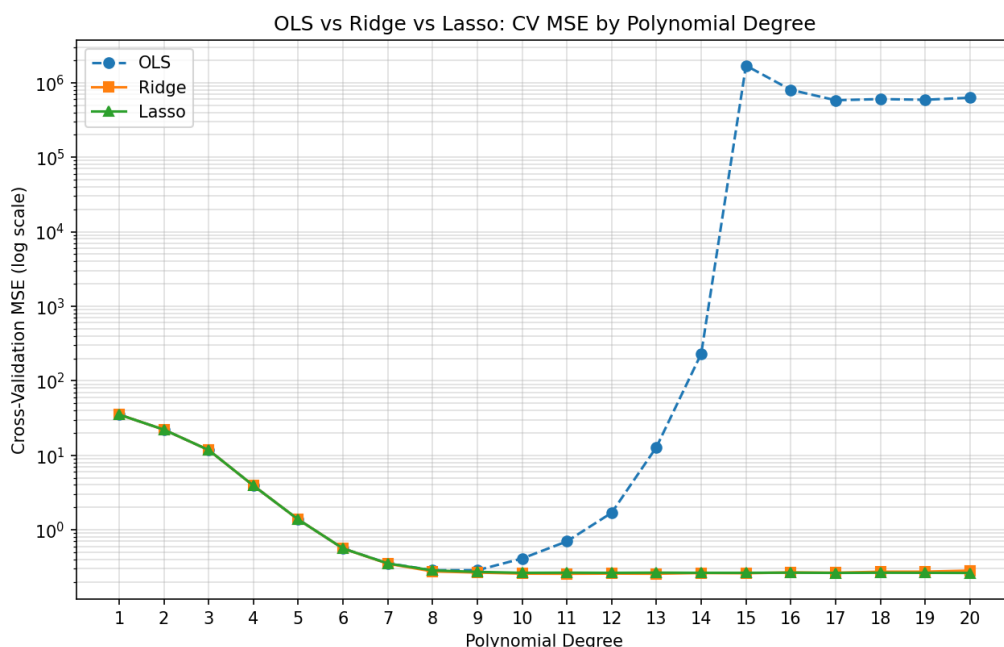


Figure 2: OLS, Ridge and Lasso cross-validation MSE for different polynomial degrees.

At lower degrees, the three methods have similar performance. As the polynomial degree increases, OLS becomes highly unstable and its validation MSE increases sharply after degree 14. Ridge and Lasso remain comparatively stable because of regularisation.

The best model was Ridge with degree 11 and

$$\alpha = 1.0$$

with

$$\text{CV MSE} = 0.257254, \quad R^2 = 0.994070.$$

After selecting the model, Ridge was retrained using the complete training dataset. The final training performance was

$$\text{MSE} = 0.169847, \quad R^2 = 0.996288.$$

The final model was then used to generate predictions for the test dataset.

5 Final Models

The selected models are summarized in Table 1.

Table 1: Summary of selected models.

Problem	Method	Degree	CV MSE	CV R^2
Phase 1	Lasso	5	0.316115	0.966992
Phase 2	Ridge	11	0.257254	0.994070

6 Conclusion

Both phases were implemented using polynomial feature expansion, standardisation, regularised regression, and five-fold cross-validation.

For Phase 1, polynomial degrees from 1 to 10 were evaluated using OLS, Ridge, and Lasso. The best model was Lasso with degree 5 and $\alpha = 0.01$, achieving a cross-validation MSE of 0.316115 and an R^2 score of 0.966992.

For Phase 2, degrees from 1 to 20 were evaluated. The best model was Ridge with degree 11 and $\alpha = 1.0$, achieving a cross-validation MSE of 0.257254 and an R^2 score of 0.994070.

The results show that increasing the polynomial degree does not always improve validation performance. OLS becomes unstable at high degrees, while Ridge and Lasso provide greater stability through regularisation. Cross-validation therefore provides a reliable way to select both the model and polynomial degree.

The final selected models achieved high R^2 values on both datasets, indicating that they capture the relationship between the input features and the target effectively.